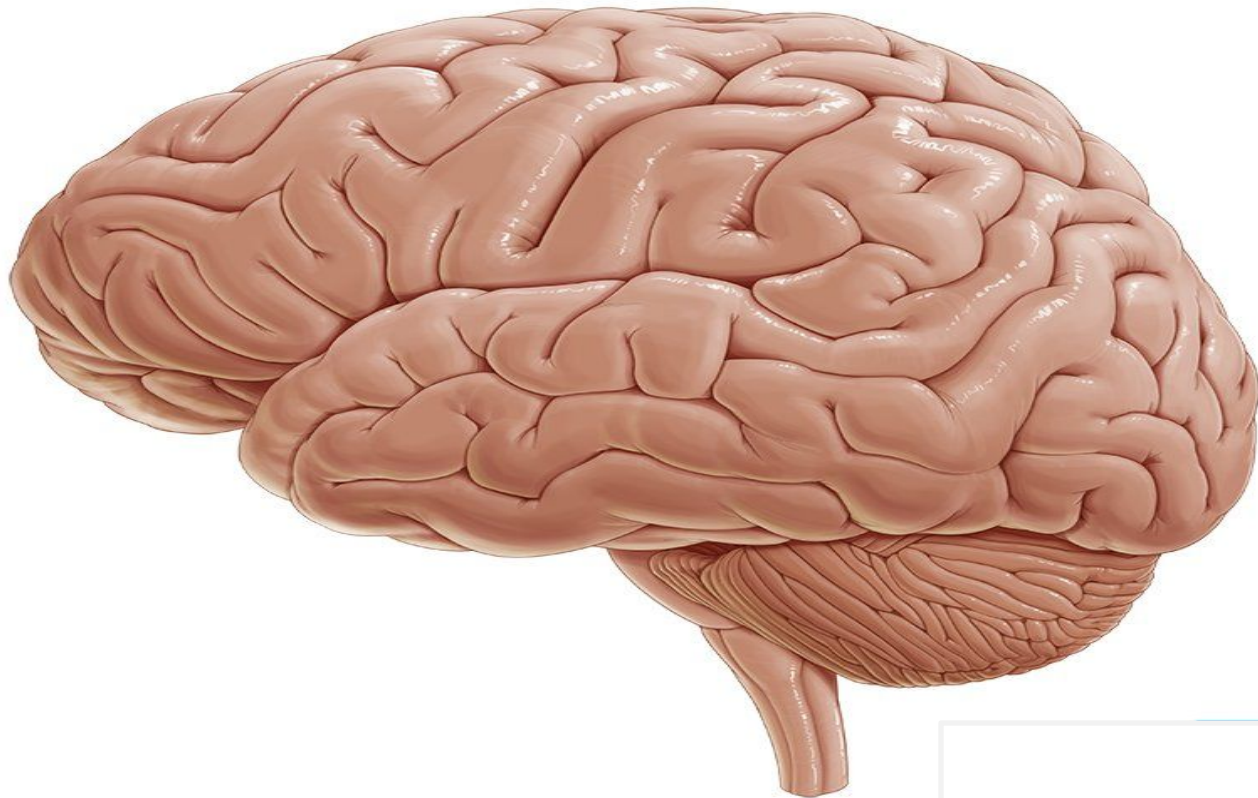


The Cerebral Hemisphere(s)

(surfaces, lobes, sulci & gyri)



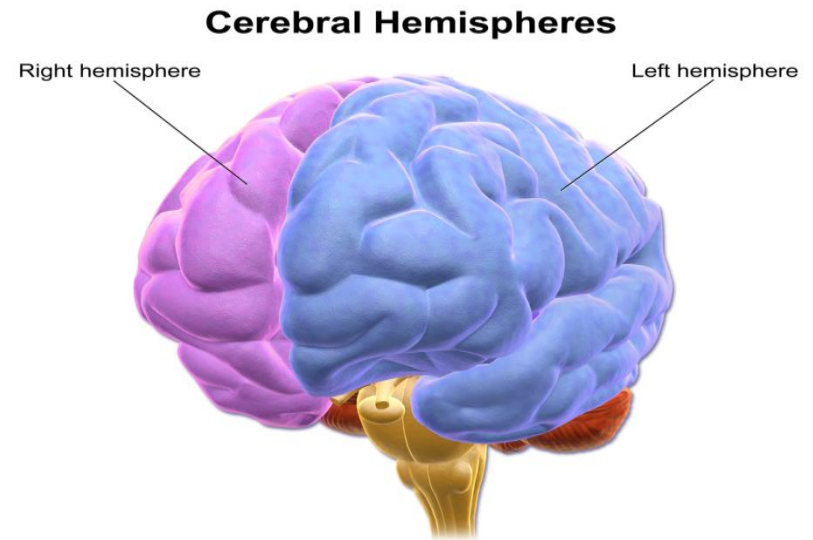
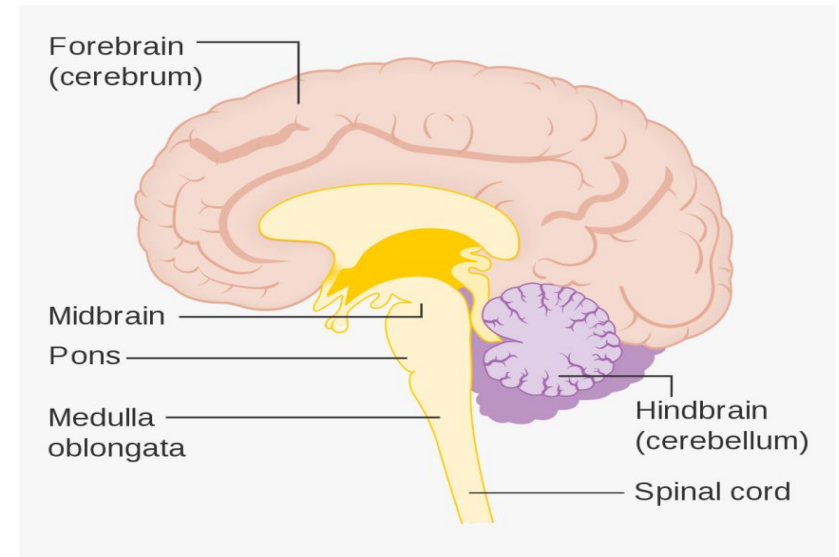
Learning objectives:

At the end of the lecture, the student should be able to:

- **Enlist parts/divisions of brain.**
- **Describe sulci and gyri of brain.**
- **Enlist borders and surfaces of each cerebral hemisphere.**
- **Identify major lobes of brain.**
- **Define important sulci and gyri on different surfaces of each cerebral hemisphere.**

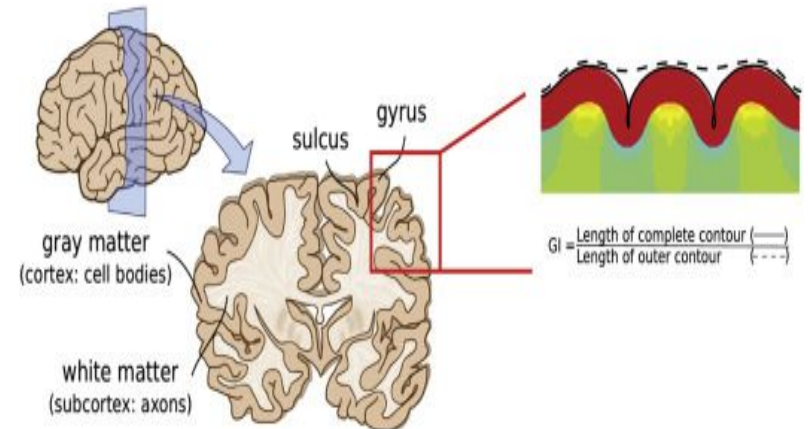
Introduction:

- Brain is conventionally divided into three main divisions:
 - Hind brain** (medulla oblongata, pons & cerebellum)
 - Mid brain**
 - Fore brain** (may be divided into **two parts**: **diencephalon**, the central core and **telencephalon**, forms the cerebral hemispheres.
- Cerebrum (with two cerebral hemispheres) is the largest part of brain.

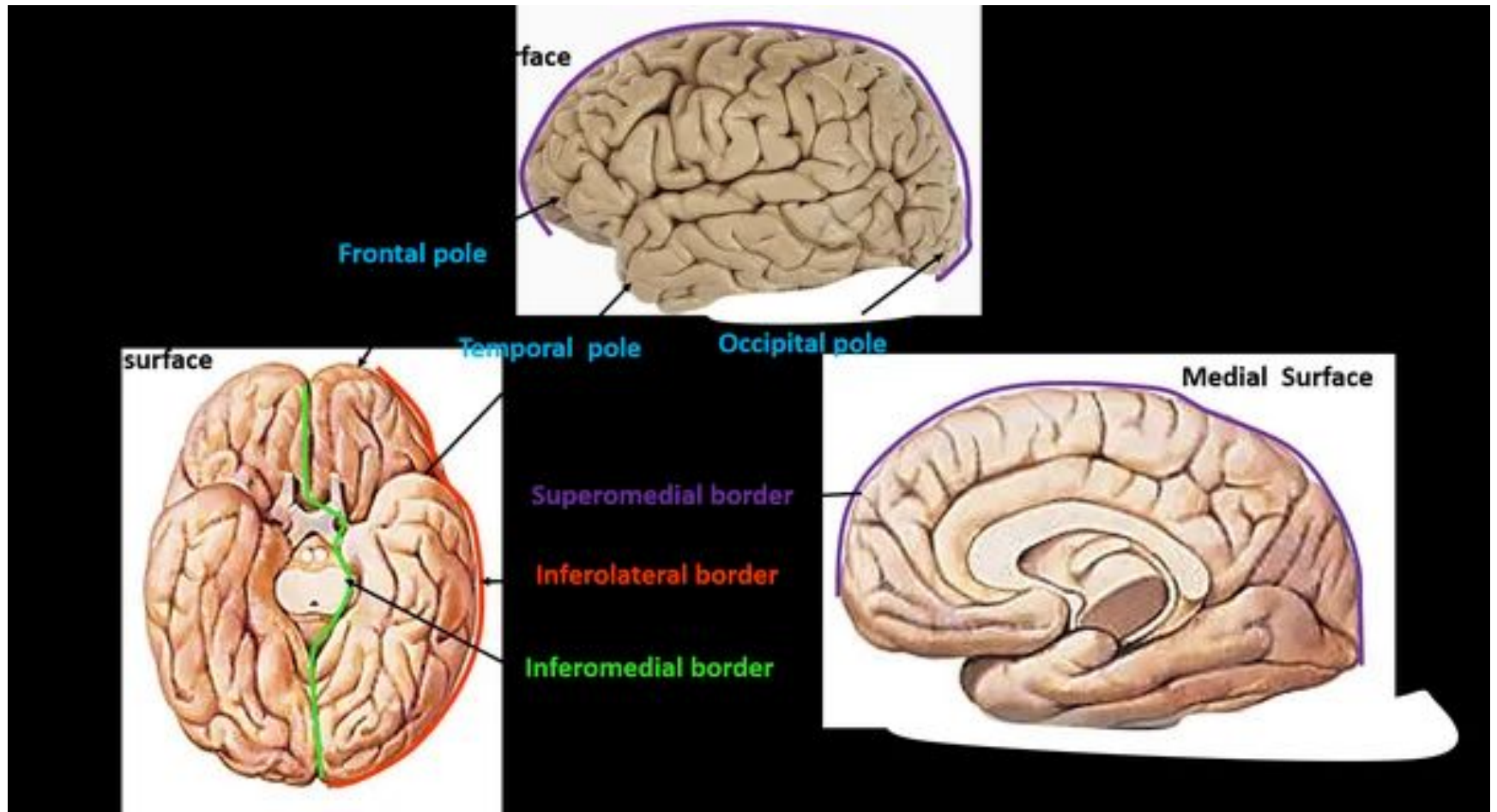


What are gyri and sulci?

- To increase the surface area of cerebral cortex, the surface of each cerebral hemisphere thrown into **folds** called **gyri/gyrus**.
- Gyri are separated by **fissures** called **sulci/sulcus**.



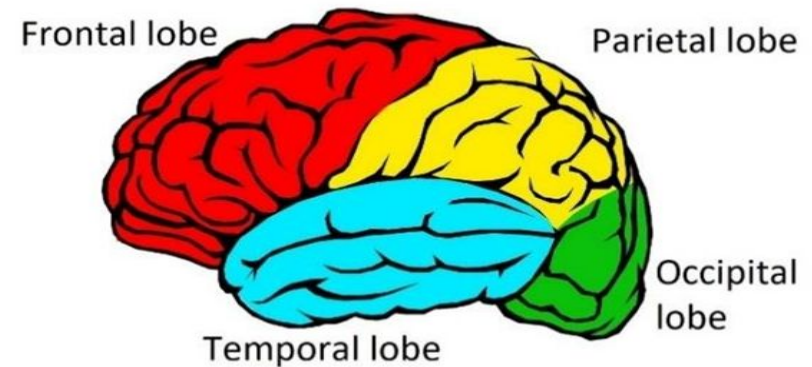
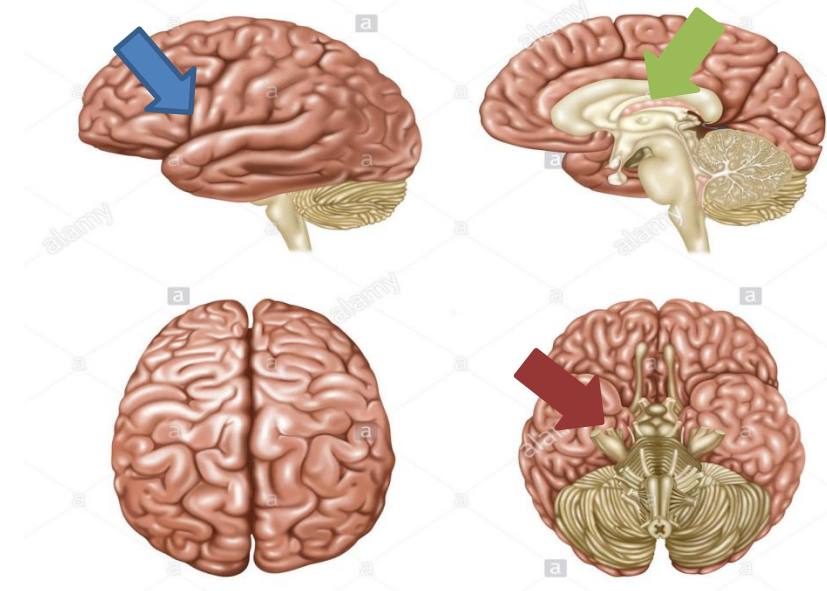
Borders:



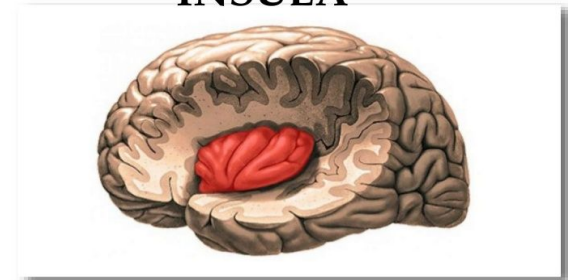
Surfaces and lobes:

- The cerebral hemispheres are separated by a midline sagittal fissure called **longitudinal cerebral fissure**
- Each cerebral hemisphere has:
 - Three surfaces: **superolateral, medial & inferior &**
 - Four lobes:
 - I. **Frontal**
 - II. **Parietal**
 - III. **Occipital**
 - IV. **Temporal &**

Insula: lies deep to lateral sulcus and can be seen by after separating the lips of lateral sulcus.



INSULA



The **main sulci** dividing each cerebral hemisphere into four lobes are:

Central sulcus

- Runs downward and forward on lateral aspect of cerebral hemisphere.
- Indents superior medial border of hemisphere 1 cm behind mid-point.
- Separated from lateral sulcus by a narrow bridge of cortex.

Lateral sulcus

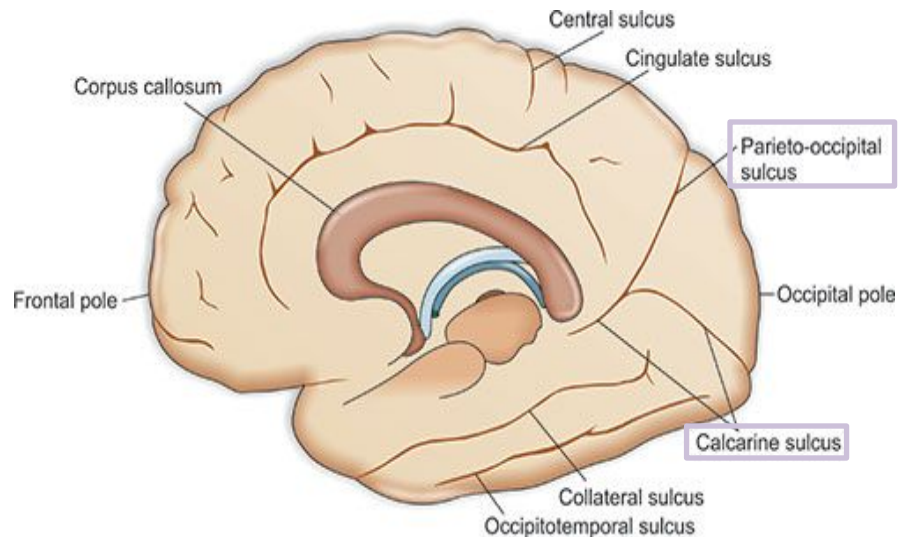
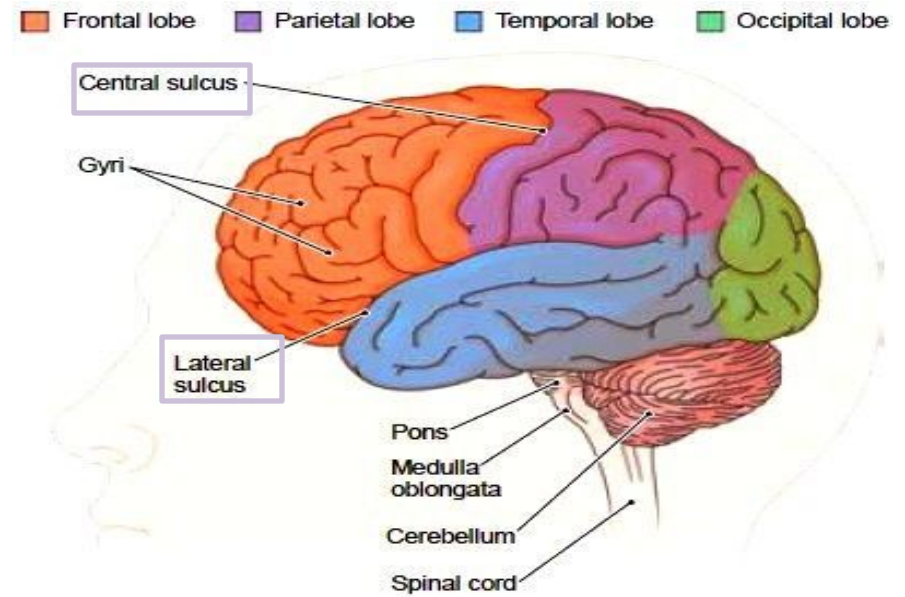
- Found mainly on inferolateral surface
- Has an anterior horizontal, anterior ascending and a posterior ramus.

Parieto-occipital sulcus

- Begins on medial margin of hemisphere 5 cm anterior to occipital pole
- Runs downwards and anteriorly to meet calcarine sulcus.

Calcarine sulcus

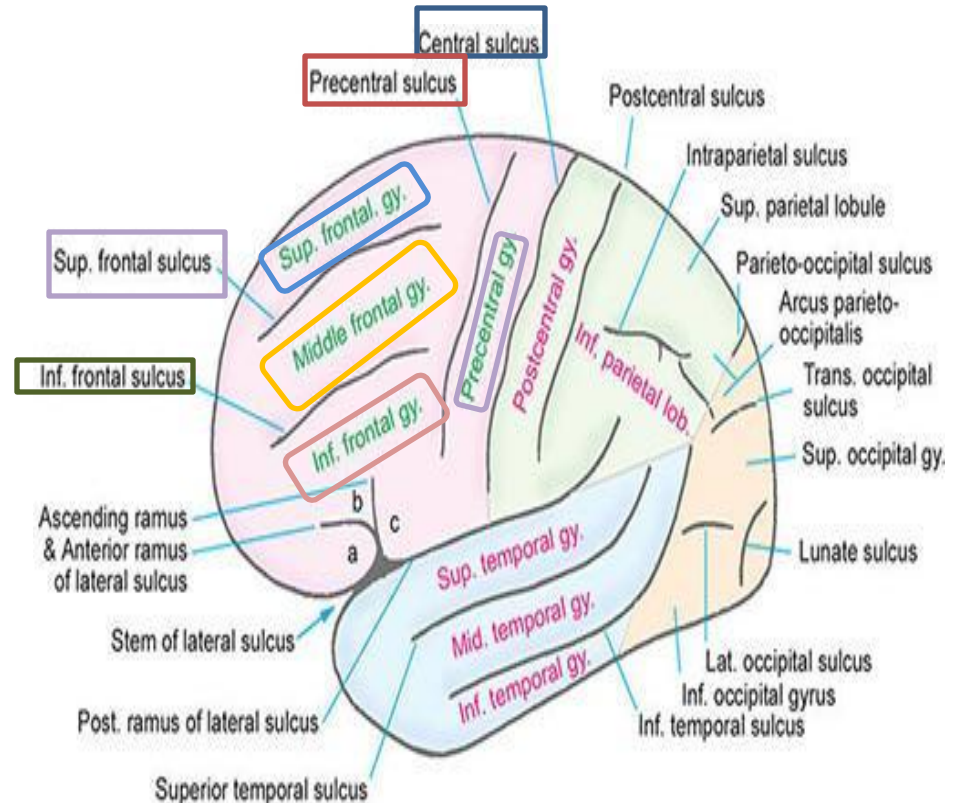
- Found on medial surface of hemisphere.
- Begins near posterior end of corpus callosum and runs backward to join parieto-occipital sulcus about midway along its length.



Superolateral surface of hemisphere:

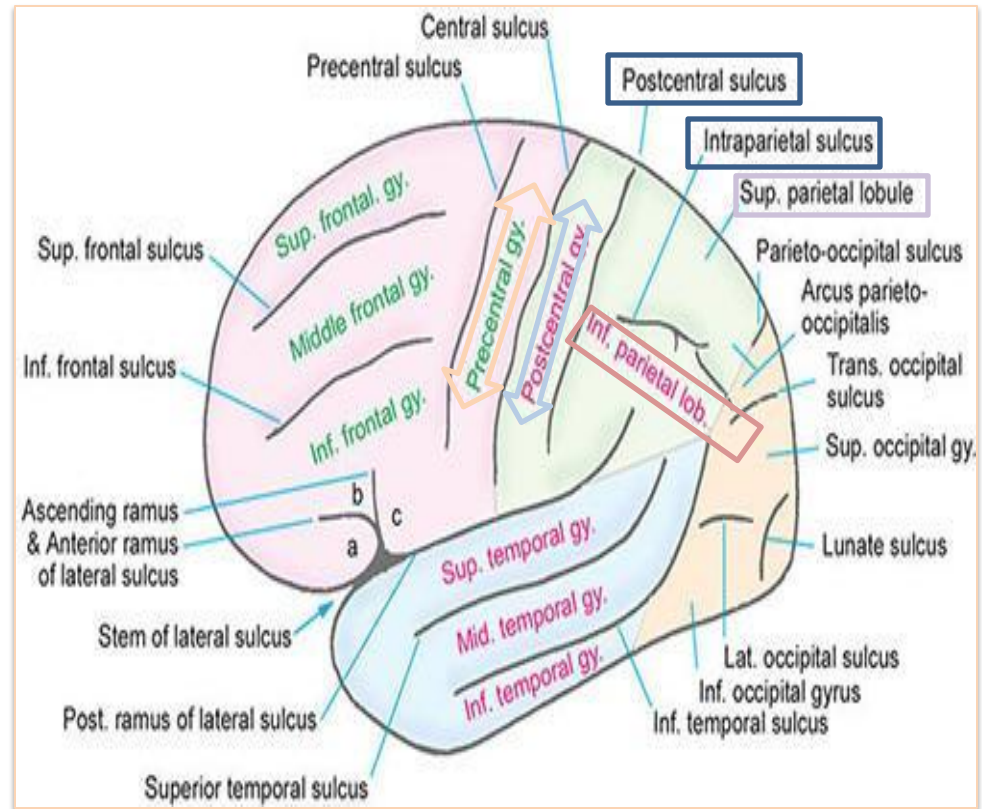
Frontal lobe:

- Lies anterior to central sulcus and superior to lateral sulcus.
- Has **three sulci**:
 - I. Precentral sulcus
 - II. Superior frontal sulcus
 - III. Inferior frontal sulcus &
- **Four gyri**:
 - I. Precentral gyrus
 - II. Superior frontal gyrus
 - III. Middle frontal gyrus
 - IV. Inferior frontal gyrus



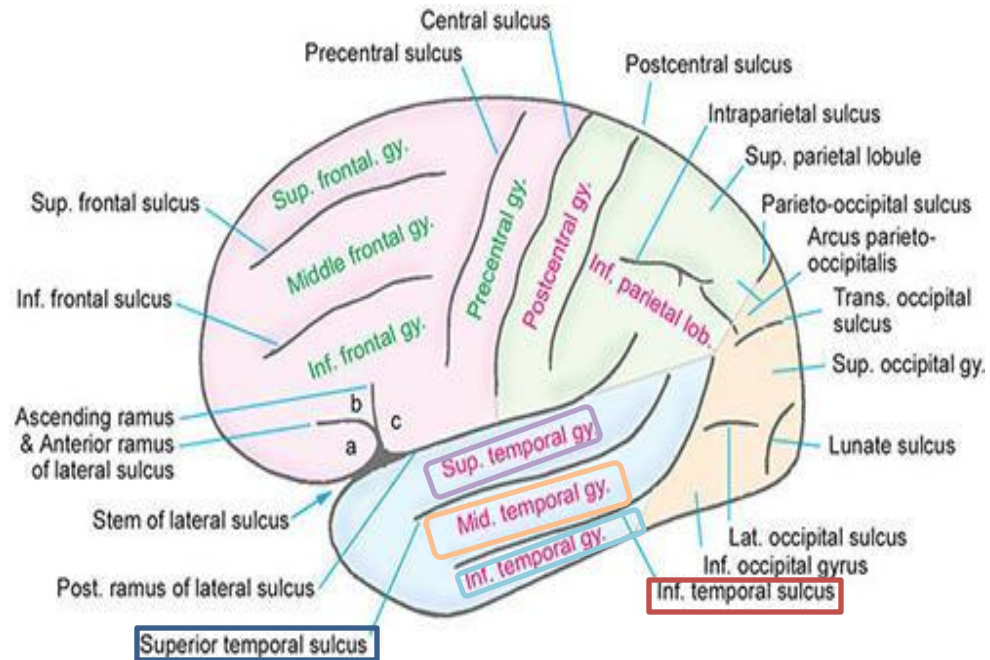
Parietal lobe:

- Lies posterior to central sulcus and superior to lateral sulcus.
- Has **two sulci**:
 - I. Postcentral sulcus
 - II. Intraparietal sulcus &
- **Three gyri**:
 - I. Postcentral gyrus
 - II. Superior parietal gyrus (lobule)
 - III. Inferior parietal gyrus (lobule)



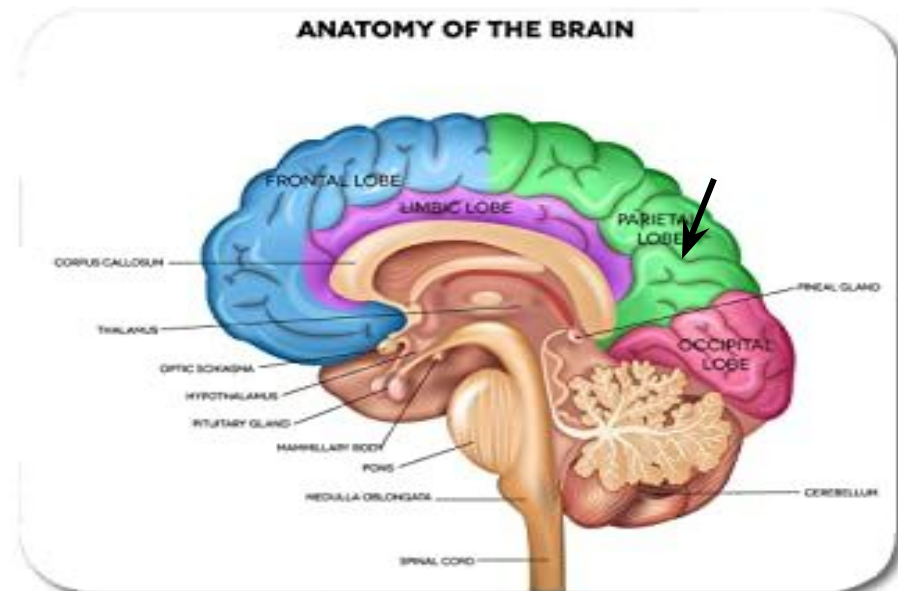
Temporal lobe:

- Lies inferior to lateral sulcus.
- Has **two sulci**:
 - I. Superior temporal sulcus
 - II. Middle temporal sulcus &
- **Three gyri**:
 - I. Superior temporal gyrus
 - II. Middle temporal gyrus
 - III. Inferior temporal gyrus



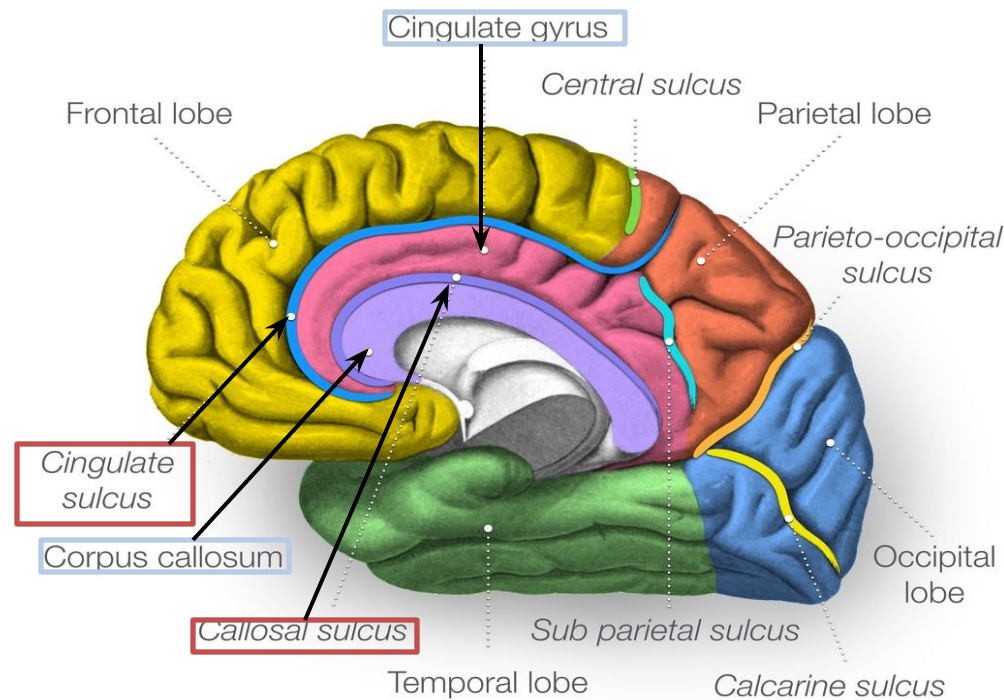
Occipital lobe:

- Smaller area lies below parieto-occipital sulcus.



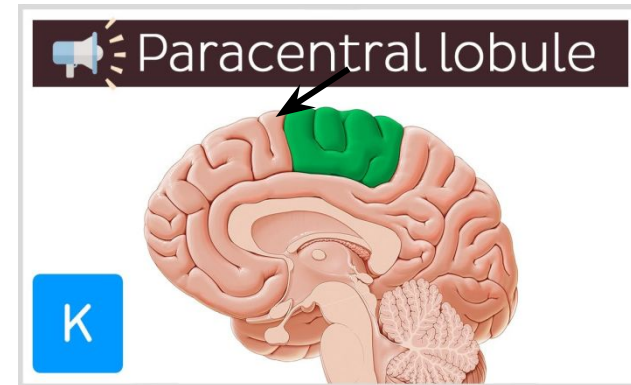
Medial surface of hemisphere:

- **Corpus callosum** is the largest commissure which connects the two cerebral hemispheres in the midline.
- **Cingulate gyrus** begins beneath corpus callosum, arches over it and ends at its posterior end.
- **Callosal sulcus** separates cingulate gyrus from corpus callosum.
- **Cingulate sulcus** separates cingulate gyrus from superior frontal gyrus.

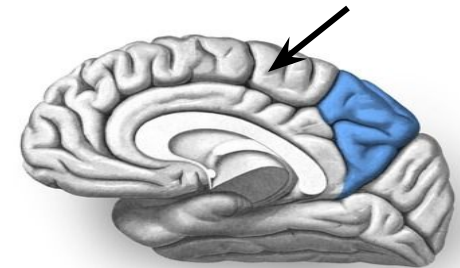


Adapted from illustration from "Sobotta's Textbook and Atlas of Human Anatomy" 1908, now in the public domain.

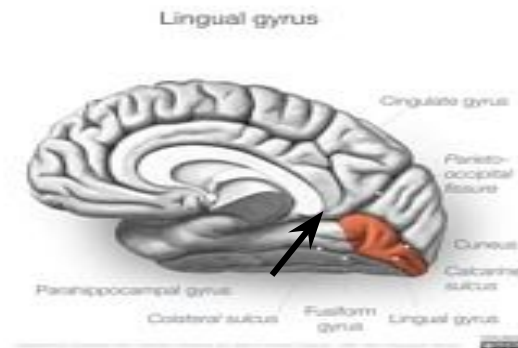
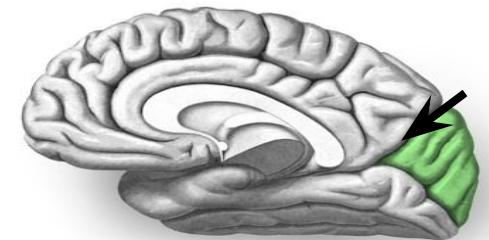
- **Paracentral lobule** lies on medial surface & is the continuation of precentral & postcentral gyri around central sulcus.
- **Precuneus** bounded anteriorly by cingulate sulcus & posteriorly by parieto-occipital sulcus.
- **Cuneus** is a triangular area bounded superiorly by parieto-occipital sulcus, inferiorly by calcarine sulcus and posteriorly by superomedial border.
- **Lingual gyrus** lies below calcarine sulcus.



Precuneus

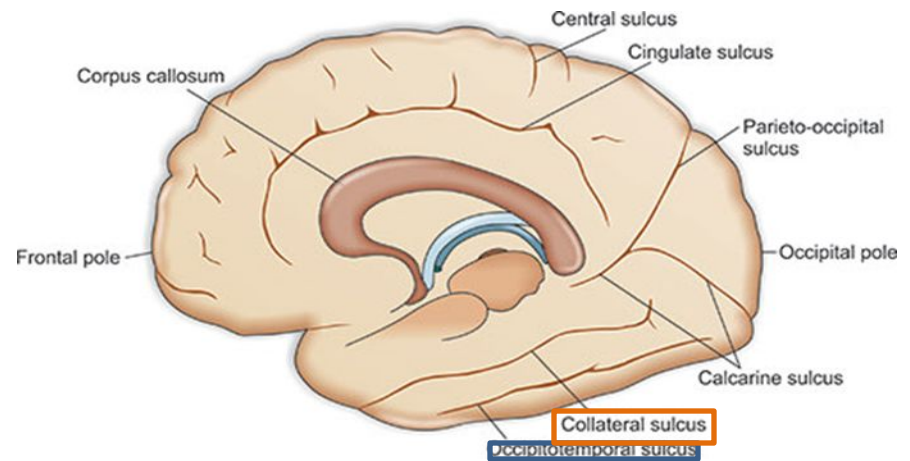


Cuneus

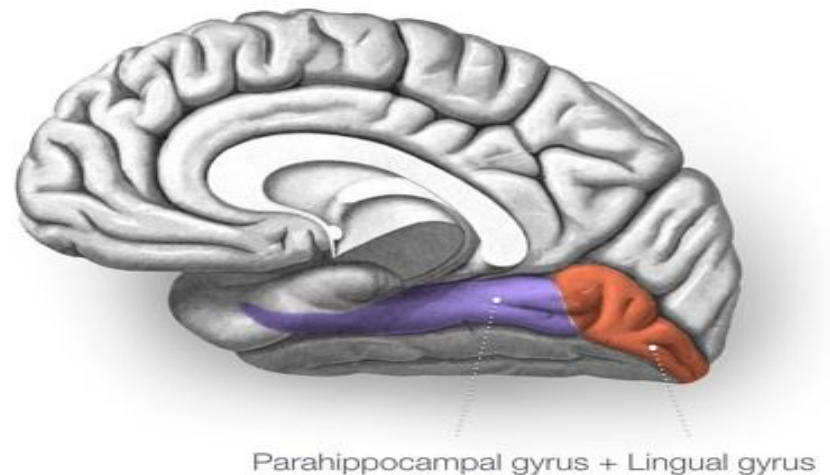


Adapted from illustration from "Solomon's Textbook and Atlas of Human Anatomy" 1908, now in the public domain.

- **Parahippocampal gyrus** lies anterior to lingual gyrus and ends as **uncus**.
- **Medial occipitotemporal gyrus** lies between collateral and occipitotemporal sulcus.

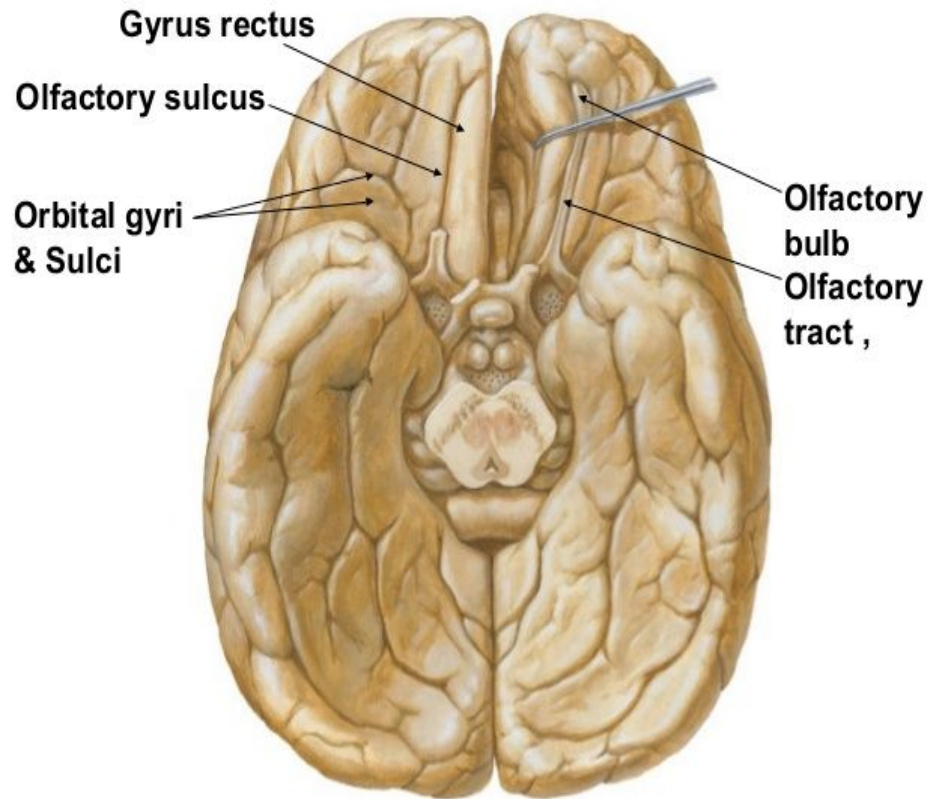


Medial occipitotemporal gyrus



Inferior surface:

- **Olfactory sulcus** lies on inferior surface of frontal lobe.
- **Gyrus rectus** lies medial to olfactory sulcus.
- **Orbital gyri** lie lateral to olfactory sulcus.



References:

- **CLINICAL
NEUROANATOMY**
Richard S. Snell
Chapter 7
8th edition
Pages # 257 -262

Thank you!

